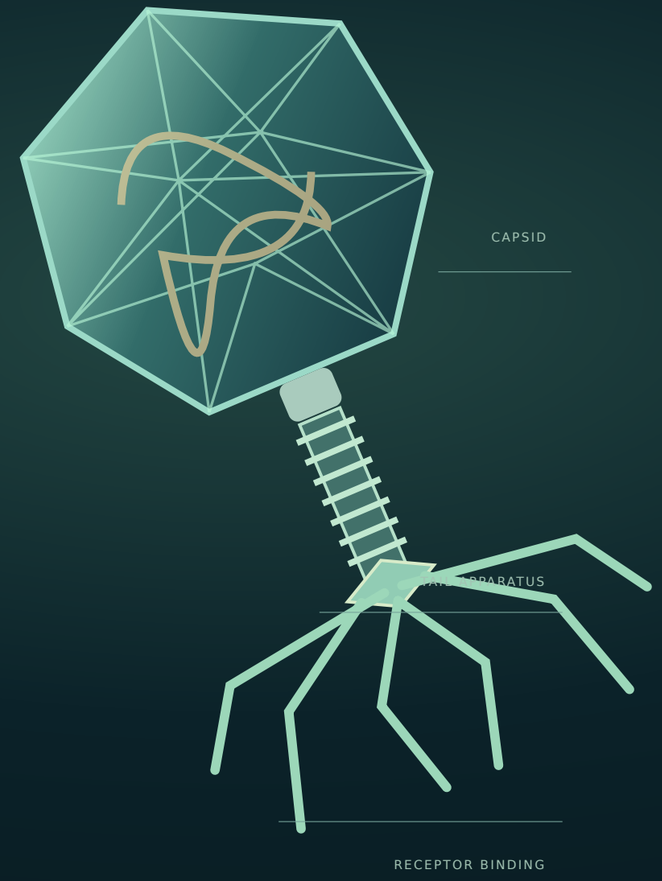
VIRUSES OF BACTERIA / A THERAPEUTIC FRONTIER

Phage *therapy.*

Turning bacterial viruses into biotechnology.



01 Match the biology.

02 Control the product.

03 Prove the benefit.

Conceptual tailed-phage illustration · Not microscopy or a scale model · Weiner et al., 2025; Aslam et al., 2020

A different route.
The same need to match.

THE DRUG PROBLEM

Antibiotic resistance

01 Drug inactivation

02 Altered targets

03 Drug efflux

Different mechanisms protect against different drugs.

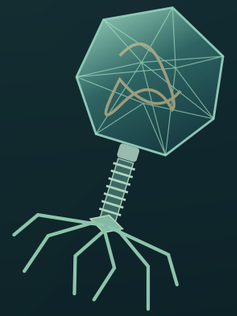
≠

THE BIOLOGICAL OPPORTUNITY

Productive infection

Recognize → Enter → Replicate

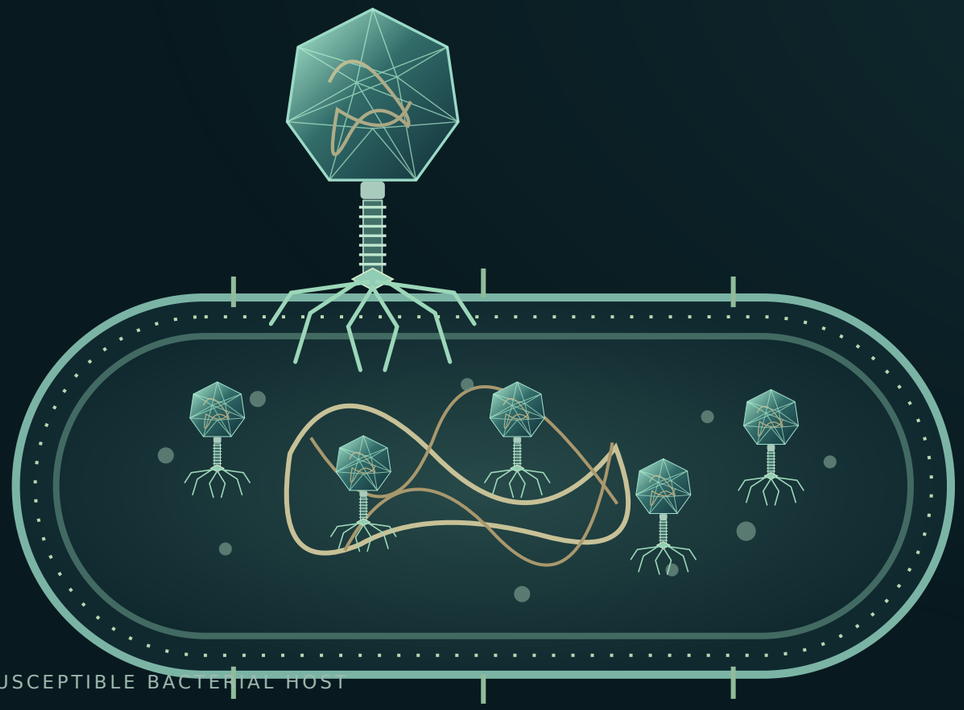
Antibiotic resistance does not establish susceptibility to a particular phage.



The therapeutic unit is a phage–isolate match.

Munita & Arias, 2016 · doi:10.1128/microbiolspec.VMBF-0016-2015 · Chan et al., 2016 · doi:10.1038/srep26717

One infection.
A new generation.



Conceptual tailed-phage model · Progeny counts are illustrative, not measured burst sizes.

01 **Attachment**
Compatible receptor recognition

02 **Genome entry**
Genetic material enters the host

03 **Replication + assembly**
Host supports progeny production

04 **Lysis + release**
New particles can infect susceptible hosts

Sequence plays automatically. Select a stage to hold it; R replays.

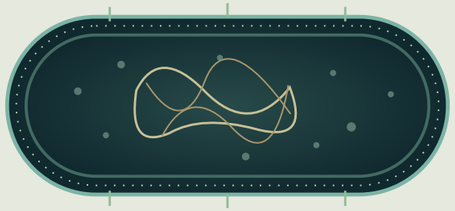
Lytic killing is the goal. Temperate phages can form prophages; amplification in a patient is not guaranteed.

Weiner et al., 2025 · doi:10.1038/s41467-025-60598-4 · Aslam et al., 2020 · doi:10.1093/ofid/ofaa389

Not any phage.
This phage. This isolate.

01

Patient isolate



Identify the infecting bacterium.

→

02

Candidate panel



Test isolate-specific activity.

→

03

Productive infection?



Spot clearing

Plaque formation

Plaques + relative plating efficiency.

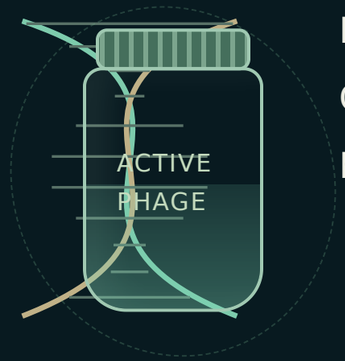
A clear spot is not proof of replication.

Conceptual assay illustrations—not experimental images, quantitative plaque counts, or clinical breakpoints.

Khan Mirzaei & Nilsson, 2015 · doi:10.1371/journal.pone.0118557 · Weiner et al., 2025 · doi:10.1038/s41467-025-60598-4

The right virus.
A suitable product.

01 / BIOLOGICAL SUITABILITY



- Lytic behavior
- Genomic identity
- Known toxin / resistance genes

Sequence and phenotype work together.

02 / FINAL-PRODUCT QUALITY

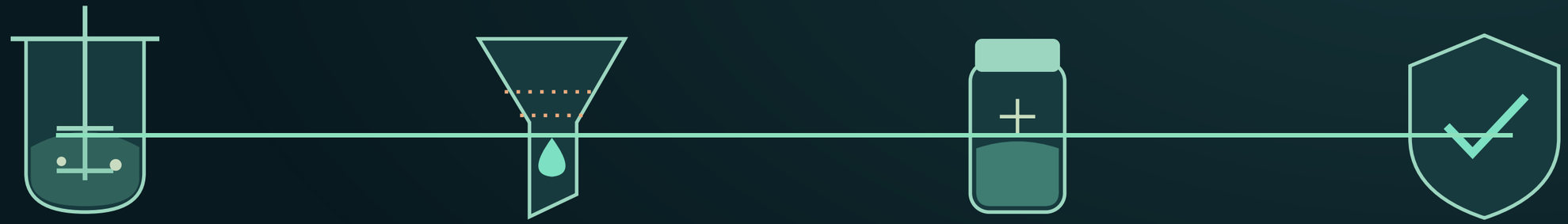
- Sterility + endotoxin
- Active titre
- Stability

Bacteria-free ≠ contaminant-free.

Screening reduces known hazards. It cannot eliminate every unknown risk.

Conceptual quality framework · Weiner et al., 2025 · doi:10.1038/s41467-025-60598-4 · Aslam et al., 2020 · doi:10.1093/ofid/ofaa389

Potency has to survive *the whole journey.*



01

Amplify

Suitable bacterial production host

02

Purify

Separate cells, debris and contaminants

03

Formulate

Compatible vehicle and delivery format

04

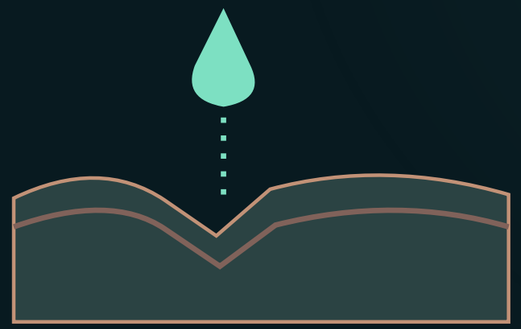
Verify

Identity, purity and active titre

Active titre measures infective particles. Detecting viral material alone does not establish potency.

Reach the infection.
Remain active.

01

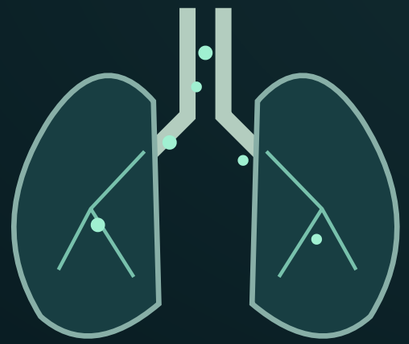


Topical

Accessible wound

PhagoBurn

02

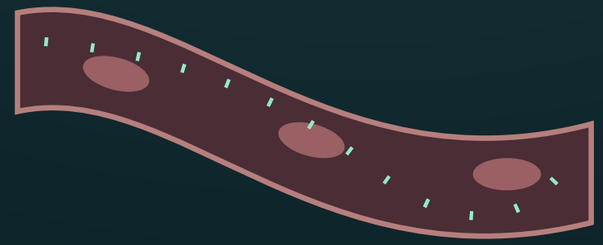


Inhaled

Lower respiratory tract

BX004-A

03



Intravenous

Systemic distribution

Personalized cases

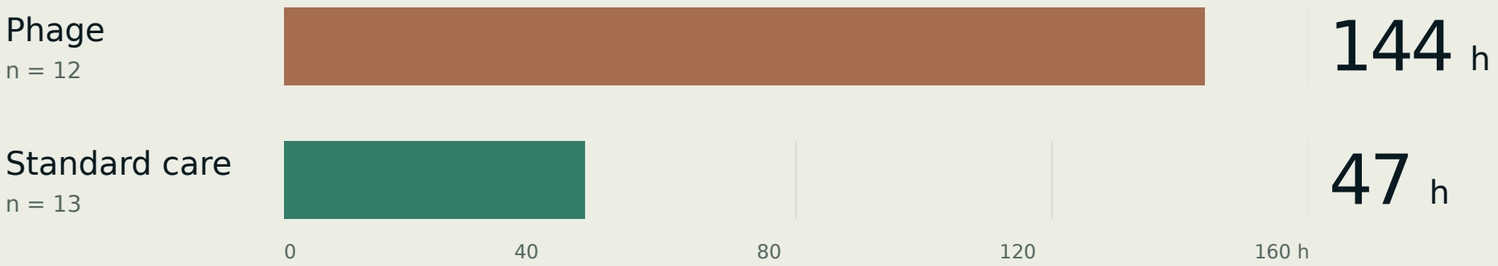
Administered dose ≠ active exposure at the site

Delivery, clearance, neutralization and bacterial susceptibility all matter. Route illustrations are conceptual, not pharmacokinetic models.

A potency gap.
An unfavorable comparison.

Pseudomonas-infected burns · topical 12-phage PP1131 vs 1% silver sulfadiazine

MEDIAN TIME TO SUSTAINED BACTERIAL-LOAD REDUCTION



HR 0.29 95% CI 0.10–0.79 P = 0.018

Longer time = slower attainment of the desired reduction.
27 randomized; 25 in the efficacy population.

A MANUFACTURING LESSON

10^6 PFU/mL

Intended concentration

↓

$\approx 10^2$ PFU/mL

Delivered after titre decline

Microbiological endpoint—not clinical cure.

Low titre does not prove that a higher dose would have worked.

Jault et al., 2019 · Lancet Infect Dis 19:35–45 · doi:10.1016/S1473-3099(18)30482-1 · Bars show reported medians; uncertainty is detailed in notes.

Delivery demonstrated.
Benefit still to prove.

Nine adults with CF + chronic Pseudomonas infection · nebulized three-phage cocktail



7 BX004-A 2 Placebo

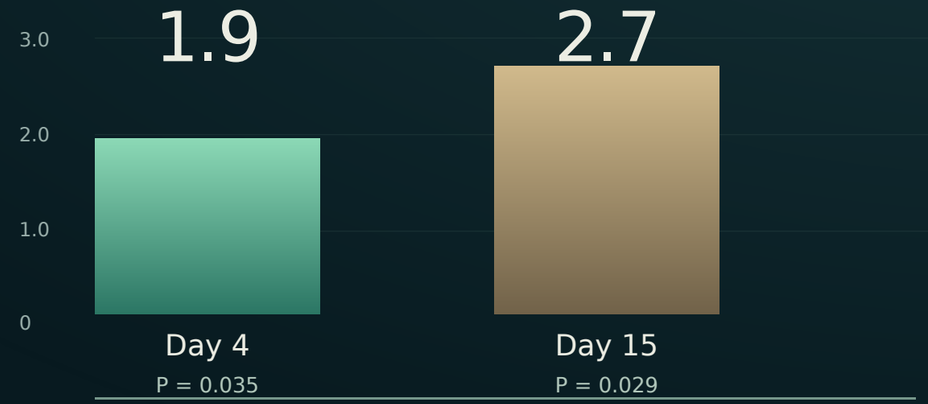
PRIMARY: SAFETY + TOLERABILITY

Endpoints met.

Phage recovery in sputum supported pulmonary delivery.

EXPLORATORY BETWEEN-GROUP DIFFERENCES IN BACTERIAL DENSITY

Reported $\log_{10}$ CFU/g lower versus placebo · selected time points



Two discrete contrasts—not a continuous response curve.
Not evidence of established lung-function or exacerbation benefit.

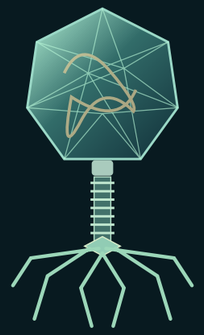
Two placebo patients. Exploratory microbiology.

2026 correction: preclinical transcriptomic sample inclusion; authors state conclusions unchanged.

Weiner et al., 2025 · doi:10.1038/s41467-025-60598-4 · Author correction, 2026 · doi:10.1038/s41467-026-70146-3

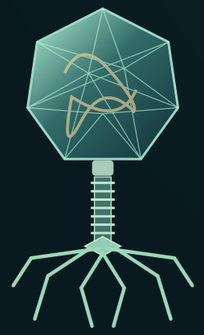
Engineering made *the match possible.*

15-year-old lung-transplant recipient · disseminated drug-resistant M. abscessus



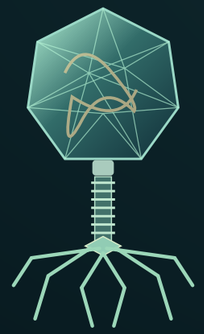
Muddy

Natural activity



ZoeJ

Repressor deletion



BPs

Lytic derivative + host-range selection

→

THREE-PHAGE COCKTAIL / IV + TOPICAL

Observed improvement.

Sternal wound closure.
Skin nodules improved.

During combined clinical care.

A feasibility demonstration. Not a controlled efficacy trial or generalizable cure.

Conceptual phage structures—not species-specific reconstructions · Dedrick et al., 2019 · doi:10.1038/s41591-019-0437-z · Activity did not generalize efficiently to other tested isolates.

Three gates.
No shortcuts.

01

Match the biology.

Active against the relevant isolate.

DEDRICK

Matching + engineering

02

Control the product.

Active product reaches the infected site.

PHAGOBURN / BX004-A

Potency + delivery

03

Prove the benefit.

Patients improve because of treatment.

CLINICAL DEVELOPMENT

Controls + meaningful endpoints

Evidence at one gate cannot substitute for the next.

Presentation synthesis—not an author-validated scoring system.

Synthesis from Dedrick et al., 2019; Jault et al., 2019; Weiner et al., 2025; Aslam et al., 2020 · Full references: slides 13–14

BUILD THE TREATMENT. TEST THE BENEFIT.

If bacterial burden falls but the patient *does not improve…*

What has the treatment demonstrated?

Match the biology. Control the product. Prove the benefit.

Clinical *evidence.*

01 **Jault P, Leclerc T, Jennes S, et al.**
(2019). Efficacy and tolerability of a cocktail of bacteriophages to treat burn wounds infected by Pseudomonas aeruginosa (PhagoBurn): a randomised, controlled, double-blind phase 1/2 trial. Lancet Infect Dis. 19(1):35–45.
https://doi.org/10.1016/S1473-3099(18)30482-1

02 **Weiner I, Kahan-Hanum M, Buchstab N, et al.**
(2025). Phage therapy with nebulized cocktail BX004-A for chronic Pseudomonas aeruginosa infections in cystic fibrosis: a randomized first-in-human trial. Nature Communications. 16:5579.
https://doi.org/10.1038/s41467-025-60598-4

03 **Weiner I, Kahan-Hanum M, Buchstab N, et al.**
(2026). Author Correction: Phage therapy with nebulized cocktail BX004-A for chronic Pseudomonas aeruginosa infections in cystic fibrosis: a randomized first-in-human trial. Nature Communications. 17:2057.
https://doi.org/10.1038/s41467-026-70146-3

04 **Dedrick RM, Guerrero-Bustamante CA, Garlena RA, et al.**
(2019). Engineered bacteriophages for treatment of a patient with a disseminated drug-resistant Mycobacterium abscessus. Nature Medicine. 25(5):730–733.
https://doi.org/10.1038/s41591-019-0437-z

Biological *foundations.*

01 **Aslam S, Lampley E, Wooten D, et al.**
(2020). Lessons Learned From the First 10 Consecutive Cases of Intravenous Bacteriophage Therapy to Treat Multidrug-Resistant Bacterial Infections at a Single Center in the United States. Open Forum Infect Dis. 7(9):ofaa389.
https://doi.org/10.1093/ofid/ofaa389

02 **Chan BK, Sistrom M, Wertz JE, et al.**
(2016). Phage selection restores antibiotic sensitivity in MDR Pseudomonas aeruginosa. Scientific Reports. 6:26717.
https://doi.org/10.1038/srep26717

03 **Kortright KE, Chan BK, Koff JL, Turner PE.**
(2019). Phage Therapy: A Renewed Approach to Combat Antibiotic-Resistant Bacteria. Cell Host & Microbe. 25(2):219–232.
https://doi.org/10.1016/j.chom.2019.01.014

04 **Munita JM, Arias CA.**
(2016). Mechanisms of Antibiotic Resistance. Microbiology Spectrum. 4(2).
https://doi.org/10.1128/microbiolspec.VMBF-0016-2015

Matching + *further trials.*

01 **Khan Mirzaei M, Nilsson AS.**
(2015). Isolation of phages for phage therapy: a comparison of spot tests and efficiency of plating analyses for determination of host range and efficacy. PLoS ONE. 10(3):e0118557.
https://doi.org/10.1371/journal.pone.0118557

02 **Wright A, Hawkins CH, Anggård EE, Harper DR.**
(2009). A controlled clinical trial of a therapeutic bacteriophage preparation in chronic otitis due to antibiotic-resistant Pseudomonas aeruginosa; a preliminary report of efficacy. Clinical Otolaryngology. 34(4):349–357.
https://doi.org/10.1111/j.1749-4486.2009.01973.x

03 **Sarker SA, Sultana S, Reuteler G, et al.**
(2016). Oral Phage Therapy of Acute Bacterial Diarrhea With Two Coliphage Preparations: A Randomized Trial in Children From Bangladesh. EBioMedicine. 4:124–137.
https://doi.org/10.1016/j.ebiom.2015.12.023

04 **Nir-Paz R, Onallah H, Dekel M, et al.**
(2025; online 2024). Randomized double-blind study on safety and tolerability of TP-102 phage cocktail in patients with infected and non-infected diabetic foot ulcers. Med. 6(5):100565.
https://doi.org/10.1016/j.medj.2024.11.018

A broader field. *Different questions.*

STUDY	POPULATION / DESIGN	INTERPRETATION
Wright · 2009	24 participants · otitis trial	Preliminary improvement; small, selected susceptible population.
Sarker · 2016	120 interim · childhood diarrhea	No clinical benefit detected; limited gut amplification; not MDR-only.
Aslam · 2020	10 heterogeneous IV cases	7 author-defined successes; uncontrolled, concurrent antibiotics; includes a previously reported case.
Nir-Paz · 2025	19 randomized · TP-102	Underpowered; no demonstrated healing superiority.

OMKO1 resistance altered an efflux-related tradeoff in vitro. Not a universal clinical resensitization effect.

Wright et al., 2009; Sarker et al., 2016; Aslam et al., 2020; Nir-Paz et al., 2025; Chan et al., 2016 · Full references: slides 14–15